

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Master of Computer Applications (MCA) /Master of Computer Applications (MCA) Lateral
Semester-V / Semester-III
University Examination November – December 2018
CA919 – Web Technologies (WT)

Date: 01.12.18, Saturday**Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. Answer to the two sections must be written in the separate answer books.
2. Answer to different questions should start on new page.
3. Figure to the right indicate full marks.
4. Make suitable assumptions wherever necessary and mention them clearly.

SECTION – I**Q-1 Do as directed.****[A] Answer the following multi choice questions.****[06]**

1. Struts2 = _____ + Struts1.
 a) Xwork b) WebWork c) Action d) Struts 1.X
2. *getText()* is provided by _____ interface
 a) Validateble b) TextProvider c) Action d) ActionSupport
3. _____ attribute is used to check the condition in *<s:if>* tag.
 a) test b) check c) value d) none of these
4. Default value for *frequency* option in *Ajax.PeriodicalUpdater()* is _____.
 a) 1 b) 2 c) 3 d) 4
5. _____ is used as a storage area for all application domain data by Struts 2 Framework.
 a) OGNL b) ValueStack c) FilterDispatcher d) none of these
6. Which attribute provides the current status of *<s:iterator>* tag.
 a) status b) first c) count d) odd

[B] State True/False.**[03]**

1. The *ActionInvocation* object represents the execution state of an action.
2. Synchronous *XMLHttpRequest* object means, it can handle multiple AJAX request at a time.
3. *execute()* method is invoked before *validate()* method.

[C] Write down the AJAX code for checking browser compatibility.**[02]****Q-2 Answer the following questions. [Any Three]****[12]**

- [A] List out classification of *Struts2 tags* and explain any one with proper example.
- [B] How many ways *Action* can be implemented? Explain with suitable example.
- [C] List out methods of *AJAX* object and explain any one method of *AJAX* object supported by *prototype.js* file with proper example.
- [D] Write a note on *Struts2 packages*.
- [E] What is the use of *AJAX*? Explain *open()* method and any two properties of *XMLHttpRequest* object.

Q-3 Answer the following questions. [Any Three]**[12]**

- [A] Explain the *MVC pattern* of *Struts2* framework.
- [B] Write a code in java (only populate method) and JSP (only tags) to show how *<select>*, *<checkboxlist>*, and *<combobox>* can be populated dynamically from a java code. (Assume proper values for each control.)
- [C] Explain how *AJAX* works with proper diagram.
- [D] What are the advantages of *Struts 2* framework.
- [E] Explain *\$()*, *\$\$()*, *\$F()* function of *prototype.js* file with suitable example.

[P.T.O]

SECTION – II

Q-4 Do as directed

[A] Answer the following multi choice questions.

[06]

1. _____ Interceptor comes first in *defaultStack*.
a) workflow b) params c) exception d) fileUpload
2. *execute()* method is provided by _____ interface.
a) Validateable b) Action c) ValidationAware d) ActionSupport
3. Web Service provides a way to achieve _____.
a) Dependency b) Interoperability c) Reusability d) None of these
4. Struts2 provides features like :
a) interceptor b) ValueStack c) OGNL d) All of above
5. *displayFormat* attribute is used in _____ tag.
a) `<s:reset>` b) `<s:select>` c) `<sx:datetimepicker>` d) `<s:submit>`
6. While creating your *custom interceptor* the real business occurs in the _____ method.
a) *init()* b) *intercept()* c) *destroy()* d) *invoke()*

[B] State True/False.

[03]

1. To get the validation message from the *property file* in the `<message>` tag, *value* attribute is used.
2. WSDL stands for Web Services Description Language.
3. *requiredstring* is considered as *built-in validator*.

[C] Assume that *list1* contains {2,1,4,3,6,5,8,7,10,9}. Write a JSP code to print **odd numbers** from *list1* using `<s:iterator>` tag. [02]

Q-5 Answer the following questions. [Any Three]

[12]

- [A] Explain *Service Oriented Architecture (SOA)* in detail.
- [B] Write a note on *Validation Framework Architecture* and explain any one build in validators with proper example.
- [C] Write down the advantages of *RESTful* web services.
- [D] Explain `<s:push>` tag and `<s:set>` tag with proper example.
- [E] What is a *web service*? Explain the *components* of web service.

Q-6 Answer the following questions. [Any Three]

[12]

- [A] Write a note on *interceptors*. and explain How *interceptors* are being fired.
- [B] Write a note on *tiles* framework? What are the advantages of *tiles* framework?
- [C] Write down the difference between *SOAP* based and *RESTful* based web service.
- [D] What is meant by *field* and *non-field* validator? Explain with examples for each validator.
- [E] Explain OGNL.
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